Highlights

PV solutions for improving the performance of interconnected microgrids

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- Smart grid
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- Aquila Optimization Algorithm
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- Practical
- Operational

PV solutions for improving the performance of interconnected microgrids

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Abstract

Energy storage enhances efficiency and facilitates the integration of variable renewables, offering increased security and flexibility to networks. Solar energy, particularly harnessed through photovoltaic panels, plays a pivotal role in reducing CO2 emissions, minimizing pollutant releases, and preserving natural resources. The recyclable nature of solar panels not only contributes to environmental sustainability but also leads to additional financial savings. This article delves into examining the weaknesses of photovoltaic panels, focusing on the voltage profile, and explores the solutions implemented to ensure their optimal performance.

I. INTRODUCTION

Rapid technological progress and intense competition drive the substantial growth in energy demand for economic development. Regional progress is closely tied to energy consumption. Renewable energy sources (RESs), especially photovoltaics, play a crucial role in advancing sustainable energy globally. The electricity generated from photovoltaics depends on sunlight intensity, and improving photovoltaic cell efficiency involves designing effective airflow channels. Mechanized cooling, as opposed to ambient cooling, further enhances efficiency. In grid-connected mode, distributed generation units regulate power using voltage and frequency; the microgrid is connected to the main utility grid when the static switch is on. Each distributed generation unit controls real and reactive power. The microgrid's interface with the main grid is crucial for maintaining synchronization among various electricity suppliers, aiming for a balanced system with high energy quality according to the country's standards.

II. STATE OF ART

A. Renewable power resources

The exploration and integration of renewable power resources into various settings have been subjects of extensive research, as highlighted by Aloo et al. [1]. They have proposed a GA-ANFIS controller for a hybrid photovoltaic-wind microgrid with BESS, confirming its functionality in meeting voltage regulation specifications amidst electricity production variations[1].

Alrasdi et al. [2] have introduced a two-thirds optimization method for voltage control in distribution networks using batteries, high-penetration wind, and photovoltaic smart inverters, addressing voltage quality issues that have hindered the integration of PV and wind energy. Experimental results have demonstrated the efficiency of the method, providing reliable voltage control for distribution networks with substantial photovoltaic and wind sources[2] .

Caballo et al. [3] have focused on the analysis and simulation of a SUIEC in San Juan de Mozarrifar, emphasizing the potential to enhance self-consumption and self-sufficiency through photovoltaic energy. The study has underscored the industrial sector's opportunity to have installed renewable energy resources and supplied energy to co-located urban areas [3] .

Timilsina [4] has analyzed the economics of distributed photovoltaic solar energy in developing countries like Bangladesh, revealing that the economics have depended on the difference in electricity production costs between implemented photovoltaic technologies and public utilities' technologies[4].

Gross [5] has emphasized the growing role of renewable energy resources in meeting future energy needs, attributing the increase in photovoltaic power generator installations to improved solar cell efficiency and manufacturing technology [5].

Bialasiewicz [6] has applied a modular simulation system to have provided cost-effective performance analysis of renewable energy systems, addressing both static and dynamic performance. New system theories and technologies have been required to have supported the development and operation of a future grid that has relied more and more on power electronics [6].

Sun et al [7] have highlighted the characteristics of power electronics as having been used in renewable energy generation and HVDC transmission systems[7] .

B. Interconnected microgrids

C. Interconnected microgrids

In the interconnected microgrid domain, Kyeremeh et al. [8] have demonstrated the segmentation of a conventional medium voltage network into a micro-network. The goal has been to convert the supply point (BSP) into a microgrid

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Nomenclature

3LT2C	*Three-level T-type converters*	*HVDC*	*High Voltage Direct Current*
AFPs	*Active Power Filters*	*IMGs*	*Interconnected Microgrids*
AHA	*Artificial Hummingbird Algorithm*	*LCE*	*Life Cycle Emissions*
BES	*Battery Energy Storage*	*LL-TSMPC*	*Low-Loss-Tolerant Sequential Model Predictive Control*
BESS	*Battery Energy Storage System*	*MA*	*Management Algorithm*
BSP	*Bulk Supply Point*	*MAs*	*Metaheuristic Algorithms*
DC	*Direct Current*	*MG*	*Microgrid*
DERs	*Distributed Energy Resources*	*MILPM*	*Mixed-Integer Linear Programming Model*
DFR	*Distribution Feeder Reconfiguration*	*PC*	*Predictive Controller*
DPNMSLS	*Delayed Proportional Normalized Mean Square Least Squares*	*PQ*	*Power Quality*
DPWM	*Discontinuous Pulsewidth Modulations*	*PSO*	*Particle Swarm Optimization*
DSC	*Deviation Settlement Charge*	*PV*	*Photovoltaic*
DSTATCOM	*Distribution Static Compensator*	*PVEG*	*Photovoltaic Energy Generation*
DVR	*Dynamic Voltage Restorer*	*SMA*	*Slime Mould Algorithm*
EDN	*Electric Distribution Network*	*SOC*	*State Of Charge*
ESOGI	*Enhanced Second-Order Generalized Integrator*	*SOP*	*Soft-Open-Point*
ESS	*Energy Storage Systems*	*SUIEC*	*Sustainable Urban-Industrial Energy Community*
GA	*Genetic Algorithm*	*SVC*	*VAr Compensator*
Ga-Anfis	*Adaptive Neuro-Fuzzy Inference System With Genetic Algorithm*	*TAC*	*Total Annual Cost*
GD-PWM	*Generalized Discontinuous Pulsewidth Modulations*	*UPQC*	*Unified Power Quality Conditioner*
GWO	*Grey Wolf Optimization*		

with rooftop photovoltaic solar power plants located in high schools along the MT grid in Ghana. The results have shown that rooftop photovoltaic power plants located in the study area can produce electricity to meet the peak demand of the power grid (8.2 MW) and have an additional 11.3 MW reinjected into the grid [8].

Examining cooperative distributed voltage mode control for isolated direct current microgrids powered by renewable energy sources, Hwang et al. [9] have quantified the economic value and environmental advantage of replacing diesel backup generators with photovoltaic microgrids plus storage for public buildings in California, which have a net-zero carbon emissions target in the electricity sector by 2045. This study has highlighted the potential savings of over a million dollars from replacing diesel backup generators with photovoltaic microgrids plus storage for Californian public buildings under various outage durations [9].

Kumar [10] has studied the control techniques for the operation of a rooftop photovoltaic solar microgrid in islanded mode. The proposed microgrid has consisted of four rooftop photovoltaic systems on various buildings, a battery energy storage system in each building, and loads. The control technique for the bidirectional DC-DC converter has been used for the battery energy storage system and has been implemented with two voltage controllers and a current controller to handle power mismatch in the proposed microgrid system [10].

In a different avenue of research, Majji et al. [11] have studied a standalone DC microgrid model that has integrated predictive control (PC) with a composite renewable energy system. PC has been developed, along with a simplified energy MA for the standalone DC microgrid. The efficiency and performance of the proposed system have been verified and validated through MATLAB/Simulink simulations and real-time simulation results using the OPAL-RT OP4510 real-time simulator under various dynamic operating conditions [11].

Moreover, Hanzei et al.[12] have worked cooperative distributed voltage mode control for isolated direct current microgrids. Using the second-order voltage dynamics of DC-DC buck converters, the authors have aimed to develop an integrated linear-quadratic regulator that has synchronized not only the output voltages but also their derivatives. The results have shown that control information has been

Optimizer	System under studied	Objective	Conclusion	Reference
GA-ANFIS	Photovoltaic-wind hybrid microgrid	To regulate its voltage amid power generation variations	Reduction rise time, settling time, overshoot, and the ability to handle non-linearities	Aloo et al. [1] (2023)
MOA	IEEE-34 bus	To provide a reliable control method for the voltage problem	Proper coordination between the BESS nd smart PV inverters	Alrasdi et al. [2] (2023)
Digital Twin simulation	PV	To simulate an urban-industrial sustainable energy community	Enhanced self-consumption	Caballo et al. [3] (2023)
Hourly data for solar irradiation	Distribution photovoltaics	To analyze the economics of stylized grid-connected distributed PV	Deployment of distributed photovoltaics	Timilsina [4] (2022)
RPM-SIM	PV arrayâinverter	To increase efficiency of solar cells	Supplying adequate power quality	Bialasiewicz [5] (2023)
Resource diversity	Renewable energy resources	To meet future energy needs	To increase importance for future energy needs	Gross [6] (2020)
MMC	Renewable energy generation and HVDC transmission systems	To support the development and operation of a future grid	High-frequency dynamics and intercoupling	Sun et al [7] (2023)

transmitted through a graph-based network, thereby reducing the volume of exchanged information [12].

In the same theme, Elseify et al. [13] have experimented the probabilistic optimal planning of multiple PV and BES systems in distribution networks to achieve an enhanced balance optimizer with time-varying load models. The results have indicated that the proposed relay low earth orbit RLEO algorithm has significantly outperformed recent optimization methods documented in the literature when it has come to addressing the challenge of optimal allocation for multiple distributed generation units [13].

The performance of microgrid operation has required hierarchical control and estimation schemes that have coordinated and monitored the system dynamics within the expected manipulated and control variables. According to Mbungu et al [14], Innovative grid technologies have come with novel concepts for two-way communication and opportunities to efficiently manage the energy flow from different independent agents [14].

Smart grid planning and control have become themes of high interest in the last years. This has been due to the presence of distributed generation, power from renewable resources, and storage systems [15] [...]. According to Minciardi et al [15], a bilevel approach for the design of a control scheme of a system consisting of a series of microgrids has been developed and applied to a case study. Preserving the frequency stability of new important blocks in smart grids such as interconnected microgrids has been a serious challenge [15].

According to Rehimi [16], interconnected microgreds would be added to smart grids as autonomous interconnected power grids [16].

Mansouri et al [17] have presented a new framework for the scheduling of microgrids and distribution feeder reconfiguration (DFR). The network studied in this paper has modified 118-bus radial distribution system consisting of 10 microgrids in which traditional loads and smart homes have been considered [17].

According to Elazab et al [18], integrating microgrids into the existing power system has increased the grid reliability and efficiency. The proposed grid-connected MG has been optimally sized using the energy management architecture in order to achieve two objectives, minimizing total annual cost (TAC) and minimizing life cycle emissions (LCE) [18].

Optimizer	System under studied	Objective	Conclusion	Reference
A solar PV Python optimization	Microgrid with rooftop solar PV plants	To convert a feeder of a BSP into a microgrid	Reduction Total losses	Kyeremeh et al. [8] (2023)
Reopt, and empirical data	PV-plus-storage microgrids	To quantify the economic value and environmental benefit with PV	Increased energy storage capacity	Hwang et al. [9], (2023)
Two-stage converter	Solar PV based microgrid	To present the control techniques with PV in islanded mode.	Power mismatched	KUMAR [10] (2024)
PMA	Autonomous DC microgrid	To develop an autonomous DC microgrid	Validated through MATLAB/Simulink and real-time simulation results	Majji et al. [11] (2022)
An integral-based linear-quadratic regulator	DC-isolated microgrids	To improve voltage synchronization in DC-isolated microgrids	Improved in terms of voltage synchronization	Hanzei et al. [12] (2023)
EO based on reinforced learning	Multiple pvand battery energy storage systems in distribution networks	To optimize the incorporation of multiple BES	Reduction in energy loss percentage	Elseify et al. [13] (2023)
Architectural control techniques	The performance of microgrid operation	To handle the dynamic performance of microgrids	Application to Interconnected microgrids/smart grids	Mbungu et al. [14] (2021)
LDM control strategies	UDM	Optimization of energy data flows	Improvement a stochastic optimization	Minciardi et al. [15] (2020)
FRM	Smart grids	To study on frequency control in AC imgs	To improve reliability, resiliency and flexibility of the future power generation systems	Rehimi et al. [16] (2020)
MILP	IEEE 118-bus test system	To propose a new framework for scheduling microgrids and distribution feeder reconfiguration	Minimization of the total operating cost	Mansouri et al. [17]
Dandelion Algorithm	Configuration of Microgrid	To optimize both the configuration and operation	Reduction the installation and operating costs	Elazab et al. [18]

D. Voltage Profile

In order to improve the voltage profile, Masood et al. [19] have proposed on the optimal placement of capacitors in the northern region of the Bangladesh transmission network to enhance the voltage profile. The solution of the MILP model has indicated that the installation of capacitors at buses 6, 7, 8, and 12 has improved the voltage magnitudes at low-voltage buses [19].

Voltage stability poses a significant challenge in power systems as they approach operating limits set by economic and environmental conditions. Vanishree et al [20] has introduced an insight into various problems being faced due to lack of voltage control in power systems and their solutions for a better Voltage-VAR balance for reliable

Optimizer	System under studied	Objective	Conclusion	Reference
Mixed integer linear programming	Transmission network for voltage profile improvement	To minimise the costs associated with capacitor banks	Optimised solution of the low voltage	Masood et al. [19] (2023)
TCPAR	IEEE 133 Bus	To maintain the voltage profile within permissible limits	A better characteristic performance and its ability for angle control and voltage impedance in transmission systems	Vanishree et al. [20] (2022)
Battery energy storage at the consumers	IEEE-33 bus radial distribution ystem	To develop a novel methodology for residential consumers utilizing battery energy storage systems	Minimized load shedding to bring the distribution network node voltages	Gupta et al. [21] (2023)
DC bus and centralized AC bus architectures	Microgrids	To enhance fault tolerance, handle system uncertainties, and improve the voltage profile	Continuity in power supply to the microgrid loads	Kishore et al. [22] (2023)
MPPT algorithm	Microgrids	To enhance the voltage profile of grid-connected PV systems in distribution grids	Maximum amount of power in grid connected PV systems	Praiselin et al . [23] (2021)

operation of the system. Voltage stability improvement and load profile management have remained the major concerns of the power system load-serving entities [20].

A demand-side management has enabled optimal strategy for active reactive power dispatch from the battery energy storage at the consumers premises to minimize load shedding have required to bring the distribution network node voltages within the specified range and optimize DSC under a frequency-linked pricing environment is investigated by Gupta et al [21]. Hybrid microgrid is emerging as a flexible and modular solution that integrates various heterogeneous energy resources and loads [21].

According to Kishore et al [22], more number of power converters present in a system results in a better voltage profile because sources and loads in microgrid are connected to the corresponding buses. In the distribution system, PV source have been the challenging one which directly disturbs the power quality issues[22].

In the studies of Praiselin et al [23], the simulation of the grid-connected PV systems has improved the voltage profile of the converter and load side[23].

E. Power losses reduction

In the field of power losses reduction, Bidyanath et al. [24] have implemented the genetic swarm optimization algorithm and a particle swarm optimization algorithm to improve the voltage profile and reduce losses using capacitors in the 132 kV Manipur transmission system. PSO and GA have provided promising results, but there exist differences between them in terms of implementation processes and the results they have generated \cite{ID-24}.

Mazaheri et al. [25] have studied a two-phase numerical study evaluating the potential of new V-shaped offset discrete ribs as artificial roughness to enhance the cooling of photovoltaic cells. The integration of nanofluid and ribbed channels has increased the merit factor parameter to 2.39, improving both electricity production and the thermal efficiency of the photovoltaic thermal system [25].

In a related vein, Moreira et al. [26] have proposed the optimal allocation and sizing of photovoltaic (PV) systems in energy distribution networks to reduce losses in power lines and minimize the economic aspects of PV implementation. The proposed method has been tested in a real distribution network at the Federal University of Paraiba (79 buses), and the results have shown an improvement in performance [26].

Shifting focus to grid integration and power quality improvement, Afsher et al. [27] have studied a photovoltaic

DSTATCOM with adaptive DC link voltage for grid integration and power quality improvement. The results have demonstrated how the PV-DSTATCOM performance has been achieved while ensuring enhanced power quality and grid integration improved [27].

Continuing in the realm of power quality enhancement, Singh et al.[28] have studied a multifunctional three-phase grid-connected photovoltaic solar energy conversion system using delayed proportional μ-law control to improve PQ. A d DMPNLMS control strategy has been presented for DSTATCOM . The control of the grid-connected PVEG system has been validated on a laboratory prototype under both balanced unbalanced load conditions and normal/abnormal grid voltage conditions uing in the realm of power quality enhancement, Singh et al.[28].

Line loss and line loss rate are important indexes to reflect the operation of the distribution network. Due to the variation of line layout, load characteristics, power supply radius, reactive compensation, and so on, the line loss rate has also been affected by the various factors mentioned above. According to Chen et al [29], the loss of the distribution network could be reduced by reasonably arranging the operation modes of the distribution network [29].

The reduction of power grid losses has been of great significance to the economic operation of the power system. Shi et al [30] have proposed a series of limit line loss analysis indicators. The line loss analysis technology based on the big data platform has been used well and has become the decisive factor for line loss management[30].

Network reconfiguration through the selection of open switch positions on the electric distribution network (EDN) have been an effective solution to reduce power loss on the EDN. However, Nguyen et al [31] have considered the optimal problem of Soft-Open-Point (SOP) placement and open switch position selection simultaneously on the EDN to reduce the power loss of the system and consider improving the voltage and current profiles [31].

The paper has proposed a DPWM strategy for shunt APFs. The suggested modulation approach identifies the current vector position in relation to the inverter voltage reference and calculates the optimal clamped duration for each phase based on switching power losses. The experiments performed by Asiminoaei et al [32] on a three-phase power inverter rated at 3 kVA, 400 V have shown that the proposed GD-PWM is a type of discontinuous modulator that adapts according to the amplitude and shape of the output current[32].

Olivares et al [33] have examined three methods of reducing distribution transformer losses. A shell-type transformer with a capacity of 500 kVA was utilized in the experiments.The test has shown that transformers have had higher losses when an overlap length of 2 cm is used [33].

Zhu et al [34] have applied the application of a coordinated static VAr compensator (SVC) device as an additional control for the reactive power (VAr) optimization problem and have analyzed the impact on system loss and voltage profile. The coordinated SVC model has had notable benefits through comparing it with VAr optimization with a general SVC model as well as VAr optimization without SVC model [34].

To relieve the power losses as well as maintain high-quality current output, a low-loss-tolerant sequential model predictive control (LL-TSMPC) has been proposed by Long [35]. In this article, an LL-TSMPC has been proposed to reduce the total power loss and improve the efficiency of the 3LT2C converter connected to the grid [35].

In the operation of the modern supply network, power quality has been a major concern because of the sensitivity of digital and modern control equipment to distortion/PQ deterioration. Two tools have been developed by Momoh et al [36], and these are an operational routine to optimize the switching of already existing DERs to reduce power quality problems using voltage deviation as an index [36].

The integration of large-scale wind power and photovoltaics into the power system will aggravate the voltage fluctuation of grid nodes. From Linlin et al [37], the multi-objective reactive power optimization method based on entropy weight and GWO algorithm has been proposed, and the multi-objective nonlinear reactive power optimization model has been established [37].

F. Power network performance optimization

In the realm of Power network performance optimization, Khan et al. [38] have employed real-time coordinated control of voltage regulation devices in a distribution network with low-to-high PV penetration. The authors have proposed an effective control approach that has optimally utilized existing voltage regulation devices without requiring new communication channels. Results have shown further optimization of the existing solution, rapidly restoring the State of Charge (SOC) to the reference level for unforeseen contingencies [38].

In a related study, Psaras et al. [39] have investigated the fault localization method for HVDC networks using a genetic algorithm on voltage profiles reconstructed in the frequency domain. The evaluation process has been facilitated by a GA regime, which, upon completion, has provided a close approximation to the fault's location [39].

Shifting focus to transmission network optimization, Linlin et al. [40] have highlighted the GWO algorithm with entropy weighting that has been introduced to solve the optimal compromise solution for multi-objective optimization of reactive power in the transmission network, taking into account network losses, voltage deviations, and static voltage stability margin. The entropy method has been employed to obtain the optimal compromise solution for reactive power optimization, and a numerical example has been analyzed using the IEEE 39-bus transmission network system. The proposed electrical reconfiguration has proven to be a highly reliable solution for effective power enhancement and loss mitigation in photovoltaic panels [40].

To illustrate the practical implications, Gholami et al. [41] have worked on a fuzzy risk-based framework for

Optimizer	System under studied	Objective	Conclusion	Reference
PSO algorithm	Voltage profile improvement and loss reduction	To calculate the optimal location and size of the capacitor.	PSO provides optimal solutions	Bidyanath et al. [24] (2023)
Two-phase nanofluid simulation	Thermal system	To assess a discrete V-shaped and staggered ribs as artificial roughness to enhance PV	Increasing the Merit parameter	Mazaheri et al. [25] (2023)
MPSO-DM	79 buses	To reduce losses	Improvement in performance	Moreira et al. [26](2022)
Reference dc-link voltage control scheme	A photovoltaic based distribution static compensator	To establish the grid integration and power quality enhancement	Guaranteeing power quality improvement and grid integration.	Afsher et al. [27] (2022)
DMPNLMS	Three-phase single-stage solar PVEG	To improve the PQ	The total harmonic distortions in the IEEE-519 standard	Singh et al. [28] (2022)
Interconnection switch	Line and transformer distribution .	To assess and optimize the reduction of line loss and line loss rate	Reduction of the loss of the distribution network	Chen [29] (2022)
MATLAB simulation	Distribution transformer	To optimize transformer limit line loss rate	To reduce the unbalance rate from 60% to 15%	Shi et al. [30] (2023)
GO	SOP	To address the optimal placement of Soft-Open-Points	To ensure the radial-topology EDN structure.	Nguyen et al. [31]
GD-PWM	Inverter current	To minimize switching power losses	To reduce the switching losses	Asiminoaei et al. [32]

energy storage system planning in PV networks. A risk-based framework has been explored to optimize the daily scheduling of charge/discharge for integrated PV and diesel generators with ESS in interconnected microgrids. Three final solutions have then been chosen from the perspective of risk-averse, risk-neutral, and risk-seeking operators among the solution set using a fuzzy concept [41].

Zooming in further, Calcabrini et al. [42] have studied Low Breakdown Voltage Solar Cells for Shade-Tolerant Photovoltaic Modules. They have analyzed how interdigitated back-contact solar cells with low breakdown voltages can contribute to improving the shade tolerance of photovoltaic modules. These findings have been supported by a four-month monitoring campaign of photovoltaic modules with different breakdown characteristics, which has shown a specific yield gain of approximately 4% in photovoltaic modules equipped with six bypass diodes [42].

Looking ahead, Zhang et al. [43] have introduced data-based voltage and var optimization control for the active distribution network while considering the reliability of the photovoltaic inverter. The study has revealed a data-driven reactive voltage optimization control strategy, considering the reliability of the photovoltaic inverter. The average junction temperature and junction temperature fluctuation of the insulated gate bipolar transistor have been introduced into the optimization objective of the model [43].

Moreover, Sanjenbam et al. [44] have studied low-voltage sensors based on a photovoltaic solar system integrated with UPQC to enhance power quality. Experimental results have illustrated superior performance under various operating conditions. Harmonics and voltage amplitudes on the load side have been limited in accordance with IEEE 1159 standards [44].

Optimizer	System under studied	Objective	Conclusion	Reference
Placing electromagnetic shields in the distribution transformer tank	Reducing losses in distribution transformers	To analyze the effects of using aluminum electromagnetic shields	Transformers had higher losses when an overlap l ength of 2 cm is used	Olivares et al. [33]
SVC model	System with 568 buses and 920 branches	To control for the reactive power (VAr) optimization problem	SVC model is benefits	Zhu et al. [34]
LL-TSMPC	Converter connected to the grid	To mitigate the losses and maintain high-quality grid current output	Reduce the power loss while maintaining a high quality of the output	Long et al. [35]
IEEE/ANSI standards	IEEE 14 bus system	To improve both transmission and distribution networks	To reduce power quality problems using voltage deviation as an index	Momoh et al. [36]
GWO and entropy weight	IEEE39 transmission network system	To evaluate the reactive power regulation performance of new energy units.	To improve the efficiency of reactive power calculation and achieve efficient global optimization	Linlin et al. [37]

Similarly, Naqvi et al. [45] have proposed a generalized control system based on an enhanced filtering integrator for improved performance of a grid-connected photovoltaic system under adverse grid voltages. This approach has limited injected currents and prevented system desynchronization during voltage sags. Increasing VCC in response to rising grid voltage has prevented the deterioration of grid PQ [45].

In the same way, Prasad et al. [46] have studied a photovoltaic solar-powered multilevel inverter with a series compensator to enhance power quality in grid-connected systems. A photovoltaic solar-integrated DVR with a new 23-level multilevel inverter has been presented to address power quality issues in the grid. The proposed system has been implemented using a dSPACE controller with a laboratory hardware prototype and a real-time OPAL-RT simulator configuration. The results have shown that the design of the proposed system has been more effective in mitigating voltage sags and has significantly improved power quality [46].

The intermittent character of the plant has caused a certain deficit in production at certain times of the day with respect to the load. The study of Sonon et al [47] has consisted in the optimization of the deficit, whose minimal value is evaluated at 42 percent, by the integration of a storage system by a lithium battery with a capacity estimated at 27 MWh. The optimization of the deficit by the method of the PSO has led to a proposal for the extension of the PV field from 25 MWp to 50 MWp [47]. .

III. System Description

A. Wild horse Otimization (WHO)

The WHO algorithm is used to solve the microgrid model. the algorithm is used to reduce investment costs (operation and maintenance) on microgrid systems, and also to reduce environmental pollution

The Algorithm For Solving

The algorithm has the following steps :

A.1. Initialize the population

The population of the WHO algorithm consists mainly of stallions and foals, with the total population set to N [48].

The initial position of each individual in the algorithm, satisfying the constraint, is given by $\square = (x_1, x_2, \ldots, x_n)$. Using the objective function as the measure, the evaluation value for each individual in the initial population is denoted as $\square = (O_1, O_2, \ldots, O_n)$. Set the proportion of stallions in the entire population as PS and randomly select stallions as leaders. Based on the number of stallions $G = \lfloor N \times PS \rfloor$ divided into G groups, distribute the remaining $N - G$

Optimizer	System under studied	Objective	Conclusion	Reference
PMU data	Voltage regulation devices in a high PV	To control efficiently the existing voltage regulation	Reduction of BESS processed energy and OLTC tapping operation	Khan et al. [38] (2023)
GA	HVDC grids.	To present a novel frequency domain for HVDC	High accuracy	Psaras et al. [39] (2023)
GWO	Wind power and PV into the power system	To evaluate the reactive power regulation performance	Improve the system node voltage quality and improve the stable operation level	Linlin et al. [40] (2022)
Fuzzy risk based framework	51-bus system	To generate electricity with less carbon dioxide emissions	An effective and time efficient classification	Gholami et al. [41] (2022)
Crystalline silicon PV modules with three bypass diodes.	PV modules	To interdigitate back-contact solar cells with low breakdown voltages	Specific yield gain of about in PV modules with six bypass diodes.	Calcabrini et al. [42] (2022)
The insulated gate bipolar transistor junction temperature	Simulation of IEEE 33 node distribution system	To solve the problems of voltage fluctuation,	Lifetime of all nodes decreases	Zhang et al. [43] (2023)
ESOGI	PV	To validate the effectiveness	Restriction of harmonics and magnitudes of load side voltages in the IEEE standard 1159	Danjenbam et al. [44] (2022)
EFGI	PV	To process the prevalent grid voltages	Increasing VCC in response to rising grid voltage	Naqvi et al. [45] (2023
DVR	Solar PV-Fed Multilevel Inverter	To enhance the power quality	Decrease in THD of grid side	Prasad et al. [46] (2022)
PSO	Solar photovoltaic power plant	To meet the increasing demand for electrical energy	To solve for the expansion coefficient and maximum stage capacity of the bank	Sonon et al. [47]

members evenly across the G groups. Then, compare the evaluation values of each group of stallions with the members in the group, and re-select the one with the better evaluation value as the group stallion.

A.2. Update the position of the group members

The algorithm employs random motion to ensure exploration of the search space. Members within each group center around the group's leader stallion, moving at different angles. To prevent concentration and enhance search efficiency, different members have different search radii. Group members update their positions using a specific formula.

$$X_i^{-j} = 2Zcos(2\pi RZ) \times \left(S_i - X_i^j\right) + S_i \tag{1}$$

where X_i^{-j} is the new position after the jth member's search in the ith group, and X_i^j is the current position of the jth member in the ith group. S_i is the current position of the ith group leader stallion, R represents a random number

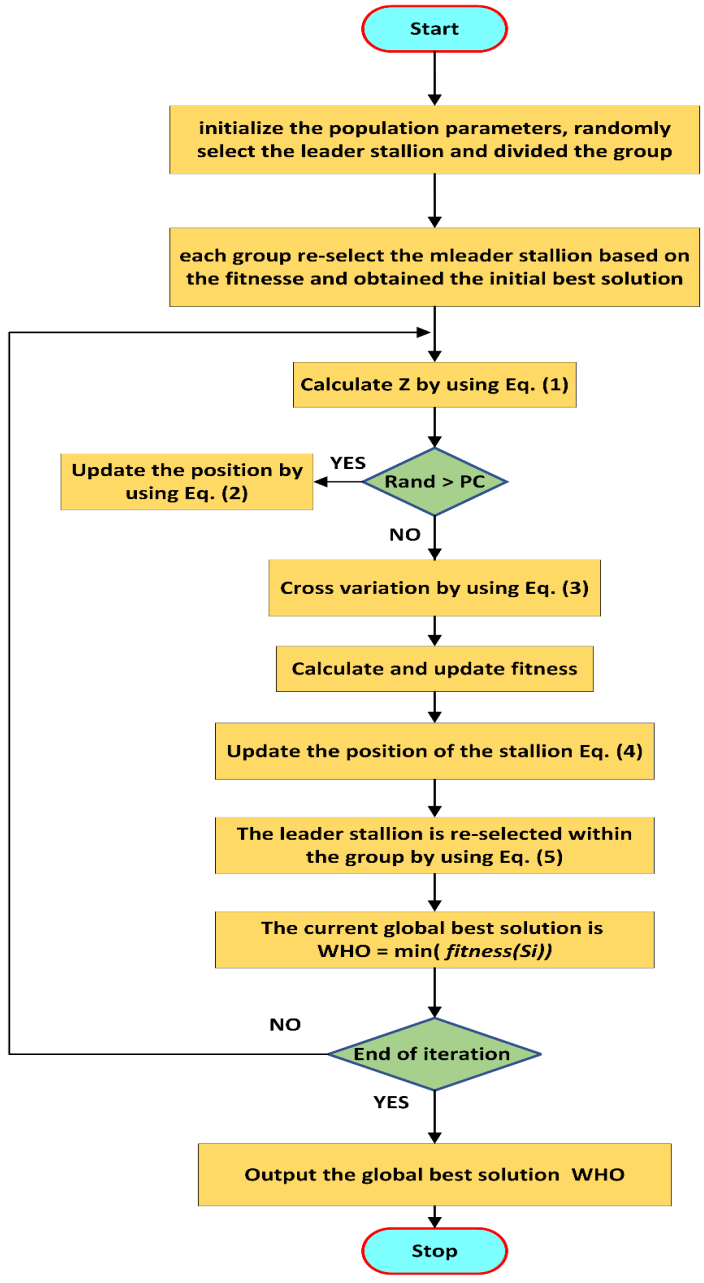


Figure 1: Flowchart of WHO [48]

and is evenly distributed in the range $[-2,\ 2]$ and Z is an adaptive structure.

$$Z = R_2 \times \mathrm{IDX} + \vec{R}_B \times (\sim \mathrm{IDX}),$$
$$\mathrm{IDX} = \begin{cases} 0, & \vec{R}_1 < IDR \\ 1, & R_1 > T\ DR \end{cases} \cdot 1DR - 1\frac{\mathrm{iler}}{\mathrm{Maxiler}} \quad (2)$$

where $\vec{R}_1$ and $\vec{R}_3$ are random vectors uniformly distributed within the range $[0,\ 1]$, R_2 represents a random number and is evenly distributed in the range $[0,\ 1]$, *iter* and *Max_iter* are the current and maximum number of iterations, respectively.

A.3. Cross-variation

To enhance the local optimization ability of the algorithm, a crossover ratio P_C is set. If the random number in $[0,\ 1]$ generated during the iterative update is less than or equal to P_C, one member from each of the two different groups, denoted as i and j, is selected. These two members are then combined to form a new member,

$$\chi_k^z = Crossover\left(X_i^m,\ X_j^n\right),\ k \neq i \neq j,$$
$$Crossover = Mean, \quad (3)$$

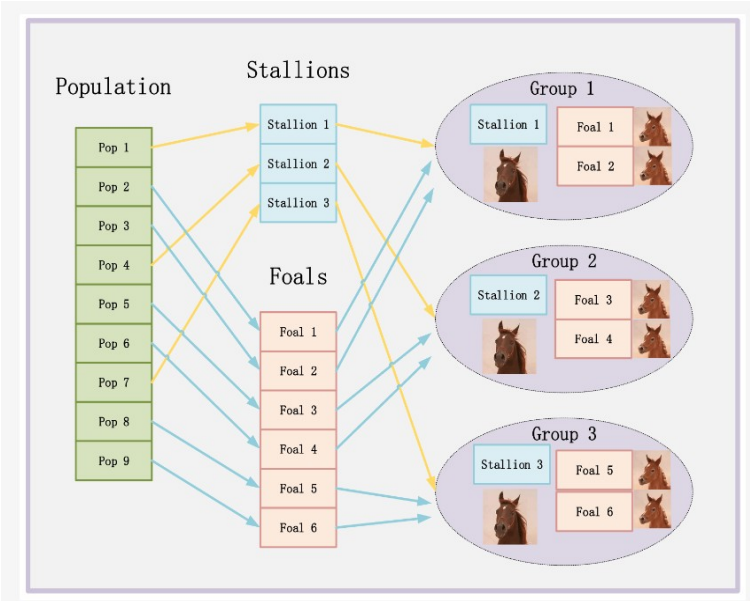


Figure 2: Representation of WHO (a)

where X_{zk} is the member z who joins the group k after cross-variation, X_{mi} and X_{nj} are members m and n of groups i and j, respectively.

A.4. Leader stallion position update

Since members of each group search for the best position around the leader stallion within the group, the stallion must compare its current position with the best global position to update it. The stallion position update formula is as follows.

$$\bar{S}_i = \begin{cases} 2Z\cos(2\pi RZ) \times (WH - S_i) + WH, & R_3 > 0.5 \\ 2Z\cos(2\pi RZ) \times (WH - S_i) - WH, & R_3 \le 0.5 \end{cases} \tag{4}$$

where $\bar{S}_i$ and S_i are the updated and current position of the leader stallion, respectively. WH is the current global best position, R and Z are the same as in (1), and R_3 is a random number within $[0, 1]$

A.5. Re-select the leader stallion within the group

After entering the iterative update, it is similarly re-select the excellent stallion to lead the group based on the evaluation value, that is, select the best evaluation value in the group as the stallion in the group, and the rest as the members of the group.

$$S_i = \begin{cases} X_i^j, & \text{fitness}(X_i^j) < \text{fitness}(S_i) \\ S_i, & \text{fitness}(X_i^j) \ge \text{fitness}(S_i) \end{cases} \tag{5}$$

Finally, the best-positioned stallion position among all stallions is compared with the current global optimal position. The flowchart of the entire algorithm is as follows [47]

B. The Slime Mould Algorithm (SMA)

The Slime Mold Algorithm (SMA) is like a computer trick that learns from how slime molds search for food. It doesn't bother with the whole life cycle of slime molds, just how they look for food. The SMA optimizer uses adjustable weights to copy the good and bad reactions slime molds have while searching for food. This leads to three different behavior types. We describe how slime molds approach things using a math model that copies their concentration mode :

$$\vec{X}(t+1) = \begin{cases} \vec{X}_b(t+1) + \vec{v}_b \cdot (\vec{W} \cdot \vec{X}_A(t) - \vec{X}_B(t)), & r < p \\ \vec{v}_c \cdot \vec{X}(t), & r \ge p \end{cases} \tag{6}$$

where $\vec{v}_b$ represents a parameter within the range $[-a, a]$, $\vec{v}_c$ is linearly decreasing from 1 to 0, t is the current iteration, $\vec{X}_b$ is individual locations with the highest currently-found odor concentrations, $\vec{X}$ is the slime mold's locations, $\vec{X}_A$ and $\vec{X}_B$ are two randomly-selected swarm individuals, $\vec{W}$ is the slime mold's weight. The value of p is modeled as follows:

$$p = \tanh|S(i) - DF| \tag{7}$$

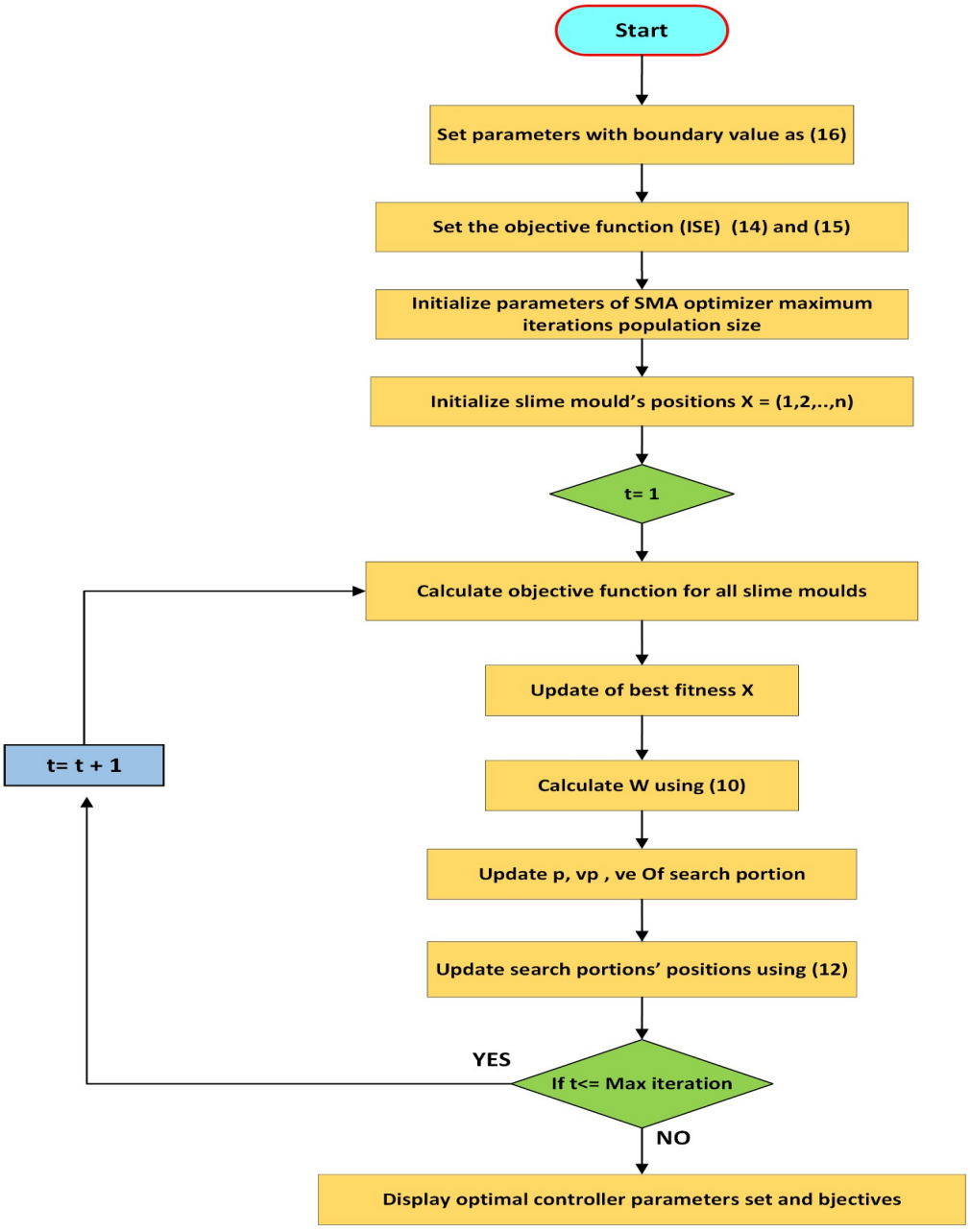


Figure 3: Flowchart of SMA [47]

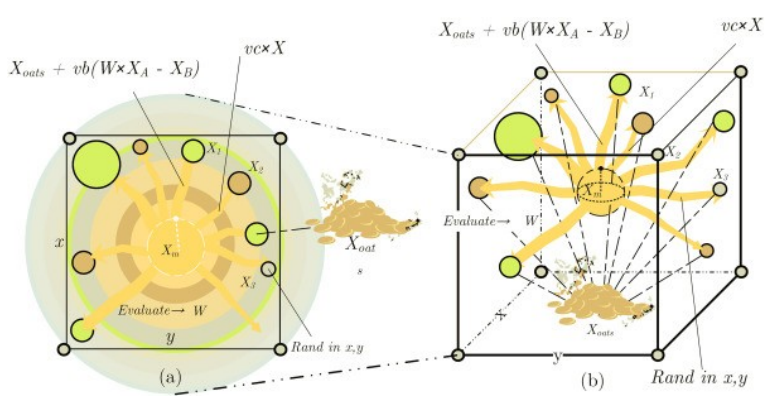


Figure 4: Representation of SMA

where, $i \in \{1, 2, \ldots, n\}$, $S(i)$ is the fitness of $\vec{X}$, DF is the best-obtained fitness value within the iterations. The value of $\vec{v}_b$ is represented as follows:

$$\vec{v}_b = [-a, a] \tag{8}$$

$$a = -\left(\frac{t}{\max_t}\right) + 1 \tag{9}$$

The formula of $\vec{W}$ is listed as follows:

$$\vec{W}(sindex(i)) = \begin{cases} 1 + r \cdot \log\left(\frac{bF - S(i)}{bF - wF} + 1\right), & \text{cond.} \\ 1 - r \cdot \log\left(\frac{bF - S(i)}{bF - wF} + 1\right), & \text{others} \end{cases} \tag{10}$$

$$sindex = \text{sort}(S) \tag{11}$$

In this scenario, the condition indicates that $S(i)$ holds a rank in the initial half of the populations. Here, r represents a random value between 0 and 1, bF stands for the best fitness achieved during the ongoing iterations, wF represents the worst fitness achieved during the ongoing iterations, and $SIndex$ signifies the sorted sequence of fitness values (arranged in ascending order for minimized value problems). The mathematical expression for updating the locations of slime molds is articulated as follows:

$$\vec{X}^* = \begin{cases} \text{rand} \cdot (UB - LB) + LB, & \text{rand} < z \\ \vec{X}_b(t) + \vec{v}_b \cdot (W\vec{X}_A(t) - \vec{X}_B(t)), & r < p \\ \vec{v}_c \cdot \vec{X}(t), & r \geq p \end{cases} \tag{12}$$

where, UB and LB are the upper and lower boundary searching range, rand and r are random values within the [0, 1]range.

B.1. application of SMA on a microgrid system

The widely-employed integral squared error (ISE) objective function is used in this work. It is expressed as following:

$$ISE = \int \sum_{i=1}^{n} (e_i^2)\, \mathrm{d}t \tag{13}$$

In a single MG, the ISE representations can be expressed for single area as following:

$$ISE = \int_0^{t_s} ((\Delta f_a)^2)\, \mathrm{d}t \tag{14}$$

And, for two-area MG, it is expressed as following:

$$ISE = \int_0^{t_s} ((\Delta f_a)^2 + (\Delta f_b)^2 + (\Delta P_{tie})^2)\, \mathrm{d}t \tag{15}$$

The minimum and maximum values for the tunable control parameters are expressed for single area MG as following:

$$\begin{aligned} K_p^{min} &\leq K_p \leq K_p^{max} \\ K_t^{min} &\leq K_t \leq K_t^{max} \\ K_i^{min} &\leq K_{i1}, K_{i2} \leq K_i^{max} \\ K_d^{min} &\leq K_d \leq K_d^{max} \\ n^{min} &\leq n \leq n^{max} \\ \lambda^{min} &\leq \lambda \leq \lambda^{max} \\ \mu^{min} &\leq \mu \leq \mu^{max} \end{aligned} \tag{16}$$

whereas, they are expressed for two-area MG as following:

$$\begin{aligned} K_p^{min} &\leq K_{p1}, K_{p2} \leq K_p^{max} \\ K_t^{min} &\leq K_{t1}, K_{t2} \leq K_t^{max} \\ K_i^{min} &\leq K_{i1}, K_{i2}, K_{i3}, K_{i4} \leq K_i^{max} \\ K_d^{min} &\leq K_{d1}, K_{d2} \leq K_d^{max} \\ n^{min} &\leq n_1, n_2 \leq n^{max} \\ \lambda^{min} &\leq \lambda_1, \lambda_2 \leq \lambda^{max} \\ \mu^{min} &\leq \mu_1, \mu_2 \leq \mu^{max} \end{aligned} \tag{16}$$

where, the limits with (f)min represent the lower limiting bounds, and the limits with (f)max represent the upper limiting bounds of the obtained optimized values for control parameters. The lower limiting bounds for control gains ($K_{\min p}$, $K_{\min t}$, $K_{\min i}$, and $K_{\min d}$) equal zero, whereas the upper limiting bounds ($K_{\max p}$, $K_{\max t}$, $K_{\max i}$, and $K_{\max d}$) equal 5 in the proposed optimization processes. The lower limiting bound for n ($n_{\min}$) is 2, whereas the upper limiting bound ($n_{\max}$) is 10. The lower limiting bounds for μ, and λ ($\mu_{\min}$, and $\lambda_{\min}$, respectively) are 0, and their upper limiting bounds ($\mu_{\max}$, and $\lambda_{\max}$) are 1 in the proposed optimization processes.

Fig. 2 [48] This figure shows how we optimize parameters using the SMA optimizer. We start by setting things up in the algorithm and defining limits as explained in equation (16). Next, we use equations (14) and (15) to establish the SMA optimizer's goal for both single and two areas. The optimizer keeps working through calculations until it reaches the maximum number of iterations.

In each round of calculations, we figure out the ISE objective for slime molds and update their best fitness X_b accordingly. Then, we use equation (12) to calculate the W value. We update the search portions (p), as well as the vectors $\vec{v}_b$ and $\vec{v}_c$. This also involves updating the corresponding positions.

Once the maximum number of iterations is reached, we get the optimized controller's parameters from the SMA optimizer.

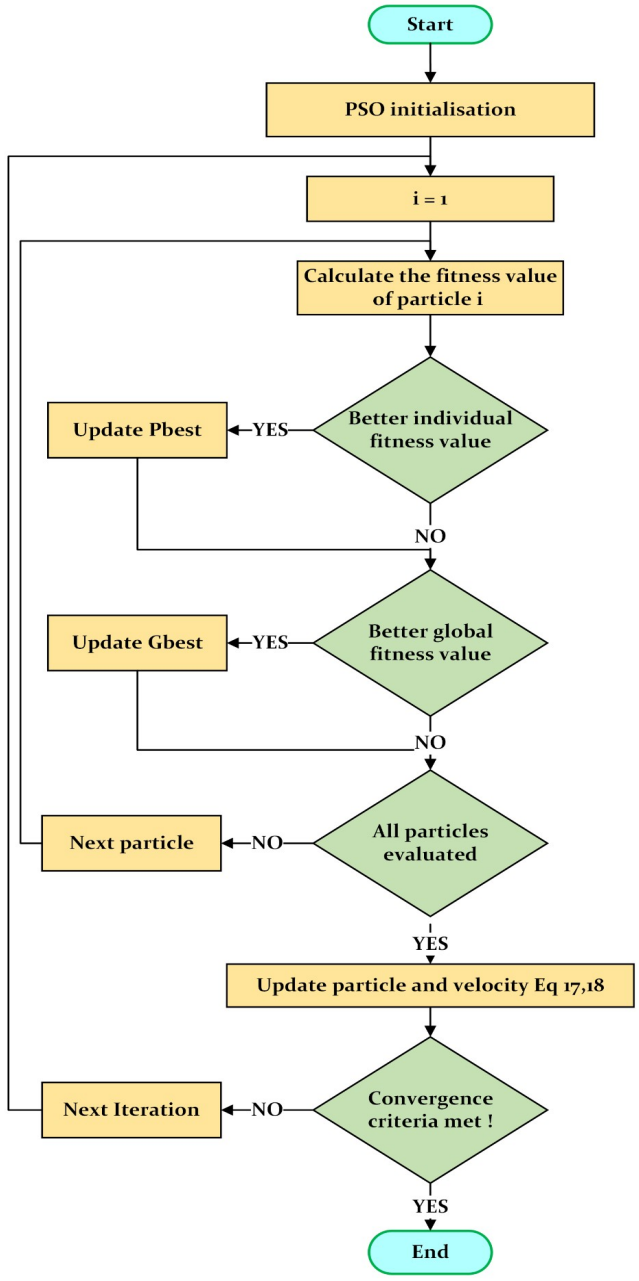


Figure 5: Flowchart of PSO (a) [49]

C. Particle Swarm Optimization (PSO)

PSO is based on the concept of a "swarm" of particles, with each particle representing a potential solution to an optimization problem. These particles move in a multi-dimensional search space in search of the best possible solution. The behavior of the particles is influenced by their own personal experience.

Here is the PSO flow chart.

The particles can gradually converge towards a global minimum [50].

- Particles: Each particle is characterized by a current position and velocity in the search space. The position represents a potential solution, while the velocity guides the particle's movement towards new solutions.
- Personal Best (PBest): Each particle retains in memory the best solution (the best position) it has found so far.
- Global Best (GBest): There is also a global best solution, which is the best solution found among all the particles in the swarm.

According to [50]:

X(t): the current position.

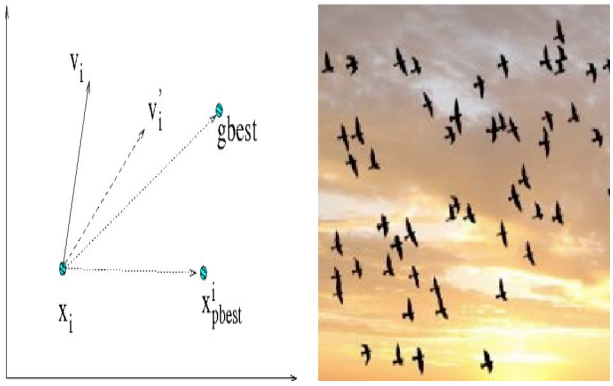


Figure 6: Representation of PSO (b)

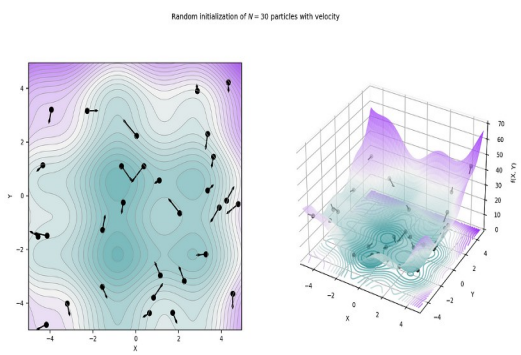


Figure 7: Representation of PSO (c)

Y(t): the personal best position.

V(t): the current velocity.

The objective of PSO is to find the configuration of particles that leads to the best evaluation of a given fitness function. Each particle represents a position in a p-dimensional REEL space . The particle's position is updated using the velocity component, calculated based on the particle's current velocity, the best position of the entire swarm, and the best individual position of the particle up to the current moment. According to [50]:

$$V^{(t)} = w * (V^{(t-1)}) + c_1 * r_1 * (Y^{(t)} - X^{(t)}) + c_2 * r_2 * (Z^{(t)} - X^{(t)}) \quad (17)$$

$$X^{(t)} = X^{(t-1)} + V^{(t)} \quad (18)$$

In the initial PSO , the velocity formula employed fixed weights: w=1, c1=c2=2, and r1, r2 are random numbers within the [0, 1] interval. In certain variations, the weight w is dynamically adjusted to decrease linearly during the algorithm's execution. For instance, it may begin at 0.9 and progressively decrease to 0.4, with updates governed by the following equation [50]:

$$w^{(t)} = w_{max} - ((w_{max} - w_{min})/T_{max}) * t \quad (19)$$

where wmax and wmin are the maximum and minimum values of w,Tmax is the maximum number of iterations and t is the current instant.

The main idea behind Particle Swarm Optimization (PSO) is to model the behavior of a swarm of particles moving in the search space to find the best possible solution. The current position of particles in PSO is represented by a vector that signifies a potential solution in the search space. Each particle also maintains an associated velocity that guides its movement in this space. The PSO algorithm evolves over iterations by adjusting the position and velocity of each particle based on its own experience and the experience of its neighborhood.

D. Aquila Optimization Algorithm (AOA)

The Aquila Optimizer (AO) was created in 2021 based on the prey-capturing behavior of Aquila. Aquila, the Latin name for the eagle (a bird of prey), is considered one of the intelligent and skillful hunters with robust feet and sharp talons, using its speed and agility to capture prey.

- High-altitude soaring with a steep dive,
- Contour flight with a short glide,
- Low-altitude flight with a gradual descent,
- And stalking and capturing prey.

The classic Aquila Optimizer (AO) primarily focuses on five significant steps, namely initialization, extensive exploration, constrained exploration, extensive exploitation, and constrained exploitation.

Initialization : During this initial phase, the solution population is randomly generated, and other parameters of the optimization algorithm are set.

Extensive Exploration essentially highlights :This phase mirrors Aquila's primary hunting technique, characterized by high-altitude soaring followed by a vertical dive. Algorithms in this phase aim to efficiently explore the search space by ascending to great heights.

Restricted Exploration :In the third phase, named restricted exploration, the algorithm emulates Aquila's second hunting technique contour flight with a short glide. Here, the algorithm maneuvers around the target by circling, akin to Aquila's precise circling around prey, as mathematically expressed in an equation.

Extensive Exploitation :The fourth phase, called Extensive Exploitation, embodies the concept of Aquila's third hunting mechanism, involving low-altitude flight with a gradual descent. In this phase, the algorithm gradually descends into the target space, approaching the intended target for optimization.

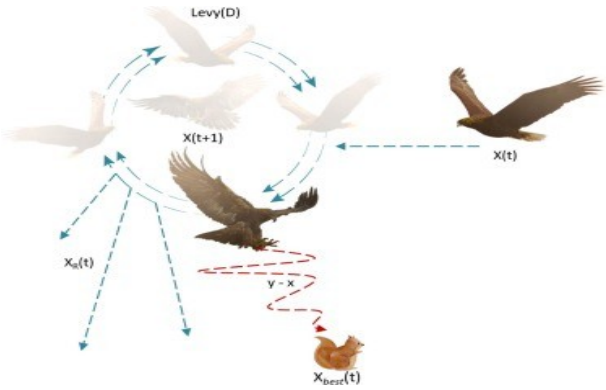


Figure 8: Representation of AHA (a)

Restricted Exploitation : Drawing inspiration from Aquila's known method of attack, referred to as the stalk-and-capture attack, the final step is termed Restricted Exploitation.

According to the article on the Optimization Droop Control Based on Aquila Optimizer Algorithm for DC Microgrid written by Aribowo et al., the AO method begins with the initialization of the candidate solution population (X). Each iteration yields the best solution.

Initialisation

$$X = \begin{bmatrix} x_{1,i} & \cdots & x_{1,\text{Dim-1}} & x_{1,\text{Dim}} \\ x_{2,i} & \cdots & \cdots & x_{2,\text{dim}} \\ \vdots & \vdots & \vdots & \vdots \\ x_{N,i} & \cdots & x_{N,\text{Dim-1}} & x_{N,\text{Dim}} \end{bmatrix} \tag{20}$$

Where, x1,i indicates the decision value (position), N is the amount of population, and Dim indicates the dimension of the issue.

$$X_{ij} = \text{Rand} \times (UB_j - LB_j) + LB_j \quad i = 1, 2, \ldots, N; j = 1, 2, \ldots, \text{dim} \tag{21}$$

- Where Rand is a disordered amount,
- LBj is the jth lower limit,
- uBj is the j the upper limit.

Step 1: Extended exploration (x1)

[51]In step 1, Aquila spots the prey zone and chooses the top chasing zone by soaring tall by a vertical. AOs is doing a lot of exploring from above soaring to determine the area of the quest room. Aquila soars with a vertical hunchback. This behavior is presented in Eq (22).

$$X_1(t+1) = X_{best}(t) \times \left(1 - \frac{t}{T}\right) + \left(X_m(t) - X_{best}(t) * \text{rand}\right) \tag{22}$$

Where xi(r+1) is the breaking of the next iteration t, produced by the initial seeking algorithm (xi).

Where

- X1(t+1) : is the solution of the next iteration t, and it is created by the first search technique X1
- Xbest(t) is the finest solution found so far
- (1- t/T) controls the exploration
- Dim is the dimension and N is the population size.

$$X_M(t) = \frac{1}{N} \sum_{i=1}^{N} X_i(t), \forall = 1, 2, \ldots Dim \tag{23}$$

Step 2: Limited exploration (x2)

[51]In Step 2, when the fugitive zone is known of an elevated height, Aquila turns the prey aim, setups the range, and then invades. This algorithm is named outline flight with a low glide invades. Here, AO narrows the exploration spot to a chosen area of aim prey in arrangement for invasion. Aquila outline flight conduct with low glide invades. This phase can be formulated in Eq (24) .

$$X_2(t+1) = X_{best}(t) \cdot \text{Levy}(D) + (X_R(t) + (y - x) \cdot \text{rand}) \tag{24}$$

- X1(t+1) is the solution of the next iteration t, and it is created by the first search technique X1
- Xbest(t) is the finest solution found
- XR(t) represents random solution belonging to [1,N],
- rand denotes random number belonging to [0,1],
- x and y presents the spiral shape in the search and is denoted using

Where x2(t+1) is Completion of the solution by the second search algorithm. D is the dimensional chamber. Levy(D) is the partition purpose of the fly levy. XR(t) is a dissimple completion grabbed in the length [1 N].

$$levy(D) = s \times \frac{u \times \sigma}{|v|^{\frac{1}{\beta}}} \tag{25}$$

$$\sigma = \left(\frac{\Gamma(1 + \beta \times \sin\left(\frac{\pi\beta}{2}\right))}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\left(\frac{\beta-1}{2}\right)}}\right) \tag{26}$$

Where s and

$$\beta$$

are a fixed value. u is a dissimple number (0-1).

Step 3: Extended exploit (X3)

[51]In Step 3 (X3), the prey spot is precisely decided, and Aquila is prepared to ground and invade, Aquila descends with an initial assail to set the response of its fugitive. The algorithm is named weak fly with the soft strike. AO uses the chosen objective zone to approach the prey and attack. This modeling can be written mathematically in Eq (27)

$$X_3(t+1) = (X_{best}(t) - X_M(t)) \times \alpha \text{ rand} + ((UB - LB) \times \text{rand} + LB) \times \delta \tag{27}$$

- T signifies the maximum number of iterations.
- UB and LB represents Upper and Lower Bound respectively,
- (α) and (δ) denotes fixed exploitation adjustment parameters.

Where x3(t+1) is solving problems in the third search method. Xbest(r) points to the reasonable spot of prey until the i-th iteration (optimal completion gained), and XM(r) is point the average grade of the current completion. alpha and delta are the value of the specified exploitation regulations (0,1).

Step 4: Limited exploits (X4)

[51]In step 4 (X4), Aquila strikes their prey on the ground suiting to their stochastic mobility. The algorithm is named steping and catching prey. AO strikes the prey in the last spot. Aquila's behavior is walking and taking prey. Step 4 can be modeled in Equation (28).

$$\begin{aligned} X_4(t+1) &= QF \times X_{best}(t) \\ &- (G_1 \times X(t) \times rand) \\ &- G_2 \times Levy(D) + rand \times G_1 \end{aligned} \tag{28}$$

Where x4(r+1) is Settlement of search solutions with the fourth algorithm. QF referred the quality purpose applied to stability the seek strategy, which is computed using Eq. (29). G1 shows the various AO movements used to track prey during elopement. G2 is a lowering value from 2 to 0, indicating the flying gradient of the AO applied to adhere prey during developement.

$$QF(t) = \frac{2 \times rand - 1}{(t - T^2)} \tag{29}$$

$$G_2 = 2 \times \left(1 - \frac{t}{T}\right) \tag{30}$$

Where QF(t) is the quality function.

$$G_1 = 2 \times rand - -l \tag{31}$$

IV. Proposed Framework

A. Cost function

$$\text{cost} = 0.25 \times \frac{P^{loss}}{P^{loss}_{init}} + 0.75 \times \frac{V^{mvm}_{init}}{V^{mvm}} \tag{32}$$

V. Simulation Results and Analysis

In Fig.6, this illustration showcases a microgrid system comprising 295 buses. The configuration correspond 294 synthetic transmission line branches and features two subsystems. The 85-bus subsystem connects to the 141-subsystem at bus-36, while the 69-bus subsystem links to the 141-subsystem at bus-34. The entire system is directly connected to the slack bus bus 1 in the 141-subsystem through a diesel generator, serving as the power source for the interconnected subsystems.

In Fig. 7 Algorithms, PSO, SMA, AO and WHO are applied to each scenario to compare current and voltage. Permitted operations are voltage limits of 0.9 min. volt and Magnitude 1 p.u. And the current limits are not mentioned. The reduction of power losses is compared to the initial losses. And the improvement of the voltage profile is analyzed in volt magnitude improvement. The voltage deviation (from an average value such as 0.95 p.u.) is also compared between the four algorithms.

A. Simulation Setup

The interconnected microgrids system, via the synthetic 295-bus, is studied under three scenarios:

- Scenario 1: each of the three connected subsystems contains its diesel generator [rated generation and load].
- Scenario 2: Only the 141 subsystem has its generator, while no generator exists in either of the 85-subsystem nor the 69-subsystem.
- Scenario 3: rated active and reactive load in whole interconnected system are elevated by 25%.

B. Results and Analysis

The table shows the results of creating a pretend bus system in Scenario 1. This pretend bus system works really well and is better than other ways of setting it up. It costs $44,625 million each year, which is less than other ways. Also, it makes more solar energy compared to other ways.

B.1. Scenario 1

Annual Cost: This is how much money is needed each year for the pretend bus system, including starting and running costs. Ours costs $44.625 million each year, which is better than other ways. The best is the Particle Swarm Optimization (PSO) method, followed by SMA, AO, and WHO.

PV Energy Production Capacity: This measures how much solar energy the pretend bus system can make in a year. Ours makes 5,992 megawatt-hours (MWh) which is more than other ways. The best is PSO, followed by SMA, AO, and WHO.

Power Factor: This is a way to see how well the system works. Ours has a power factor of 0.9319, which is better than other ways. The best is PSO, followed by SMA, AO,

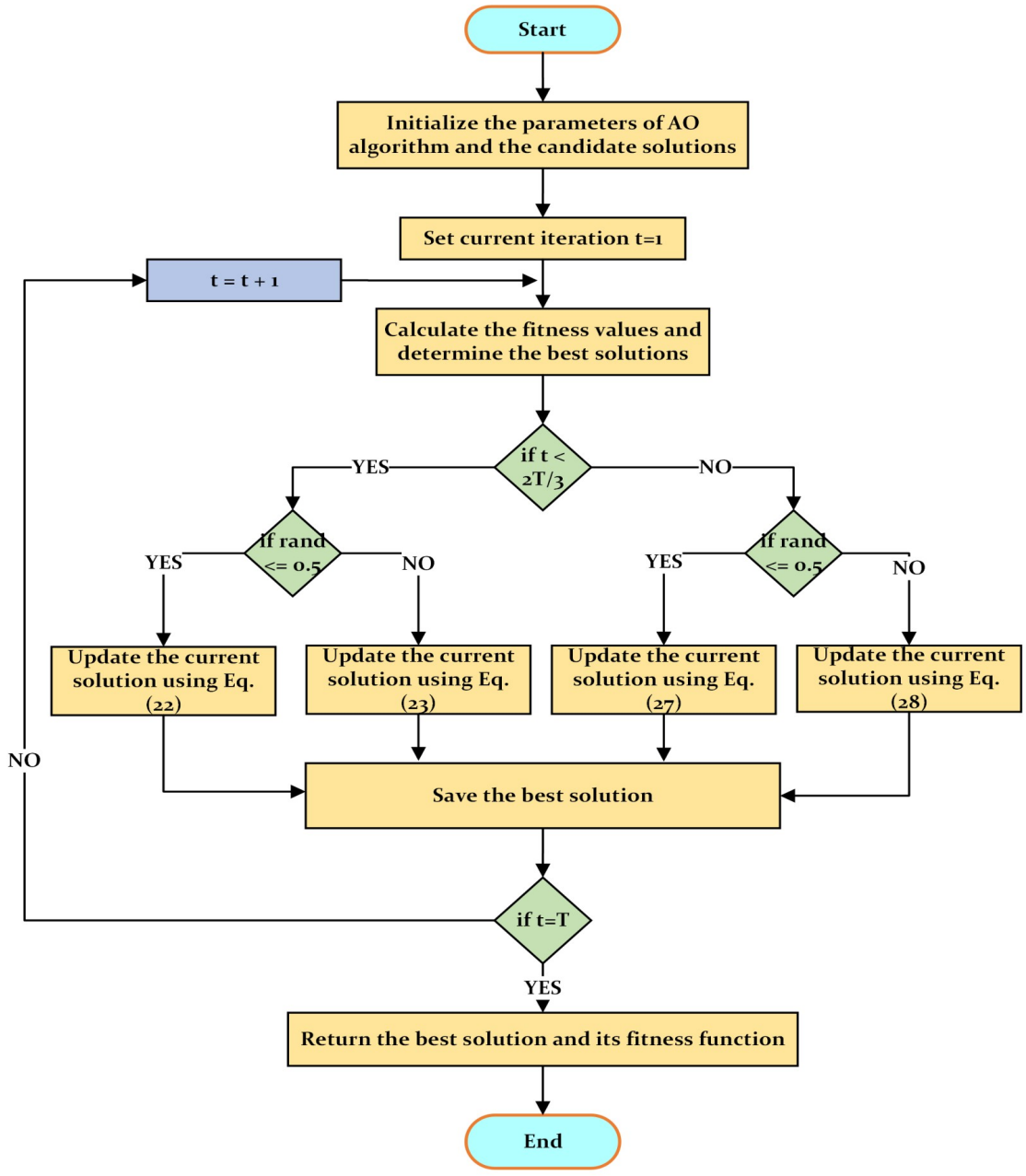


Figure 9: Flowchart of AHA (b) [51]

and WHO.

In conclusion, the table shows that using the PSO method is the best for making solar energy in our pretend bus system. It costs the least, makes the most energy, and works the best.

Here's a detailed analysis of the results of the best three optimizers:

Convergence Curve: PSO: The convergence curve for PSO shows that it reaches the optimal solution quickly.

SMA: The convergence curve for SMA is slightly slower than PSO.

AO: The convergence curve for AO is closer to SMA.

WHO: The convergence curve for WHO has the slowest slope among the four optimizers.

Average Voltage Profile: PSO: The average voltage profile for PSO shows a stable and consistent voltage across the nodes, ranging from 20 to 100.
SMA: The average voltage profile for SMA has a similar trend to PSO, with a slight fluctuation between 40 and 200.

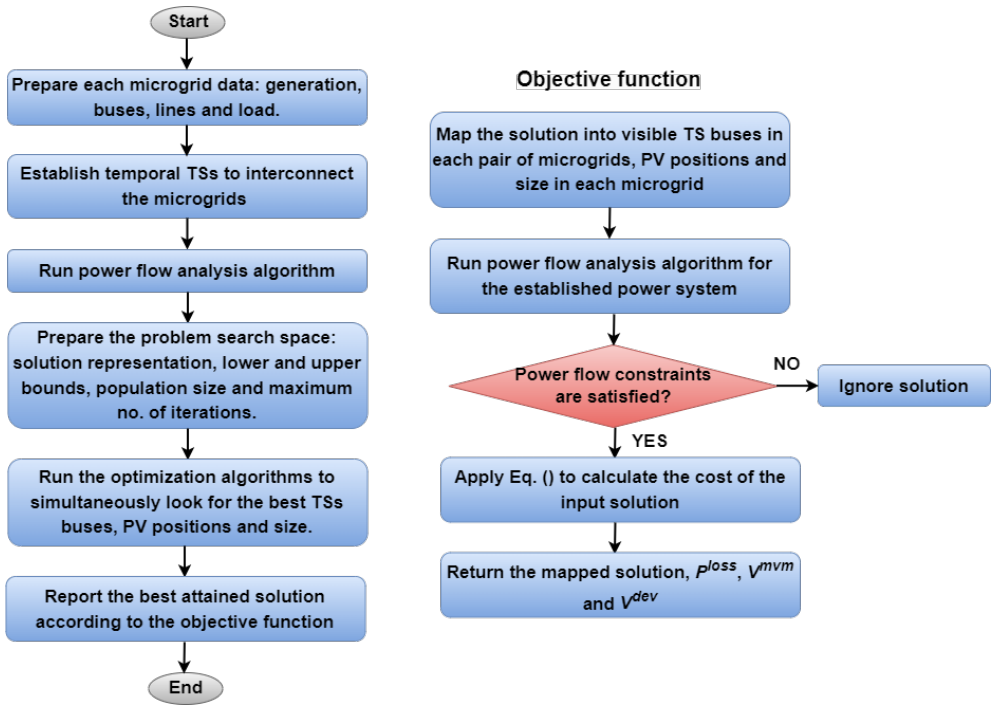


Figure 10: Flowchart of the proposed approach with the steps of the applied objective function.

AO: The average voltage profile for AO is slightly lower than PSO and SMA, ranging from 0,06 to 50.
WHO: The average voltage profile for WHO has the lowest voltage among the four optimizers, ranging from 0,06 to 300.

Best Voltage Profile: PSO: The best voltage profile for PSO shows the highest voltage across the nodes, reaching 0.96 at some nodes.
SMA: The best voltage profile for SMA is slightly lower than PSO, with a maximum of 0,85.
AO: The best voltage profile for AO is closer to SMA, with a maximum of 0,88.

WHO: The best voltage profile for WHO has the lowest voltage among the four optimizers, with a maximum of 0,36.

Overall, the figure suggests that PSO is the most effective optimizer for scenario 1, achieving the fastest convergence speed, the best average voltage profile, and the best best voltage profile.

B.2. Scenario 2

The table presents findings from a 295-bus power system under scenario 2

Four photovoltaic (PV) placement algorithms PSO, SMA, AO, and WHO are compared, evaluating cost, installed power, power loss, average power factor, and power density.

Cost : WHO algorithm minimizes cost most effectively, followed by AO, SMA, and PSO.

Installed Power : PSO algorithm yields the highest installed power, succeeded by SMA, AO, and WHO.

Power Loss : AO algorithm demonstrates the least power loss, followed by WHO, PSO, and SMA.

Average Power Factor : AO algorithm achieves the highest average power factor, trailed by WHO, SMA, and PSO.

Power Density : PSO algorithm attains the highest power density, followed by SMA, AO, and WHO.

Conclusion : In summary, WHO algorithm proves superior for PV placement in scenario 2, excelling in cost, power loss, and average power factor. AO algorithm also performs well, balancing cost, power loss, and power factor effectively.

The curve represents the average voltage profile for three PV placement algorithms: PSO, AO, and WHO.

The curve illustrates that the average voltage profile of the WHO algorithm is the highest. This indicates that the WHO algorithm is the most effective in maintaining

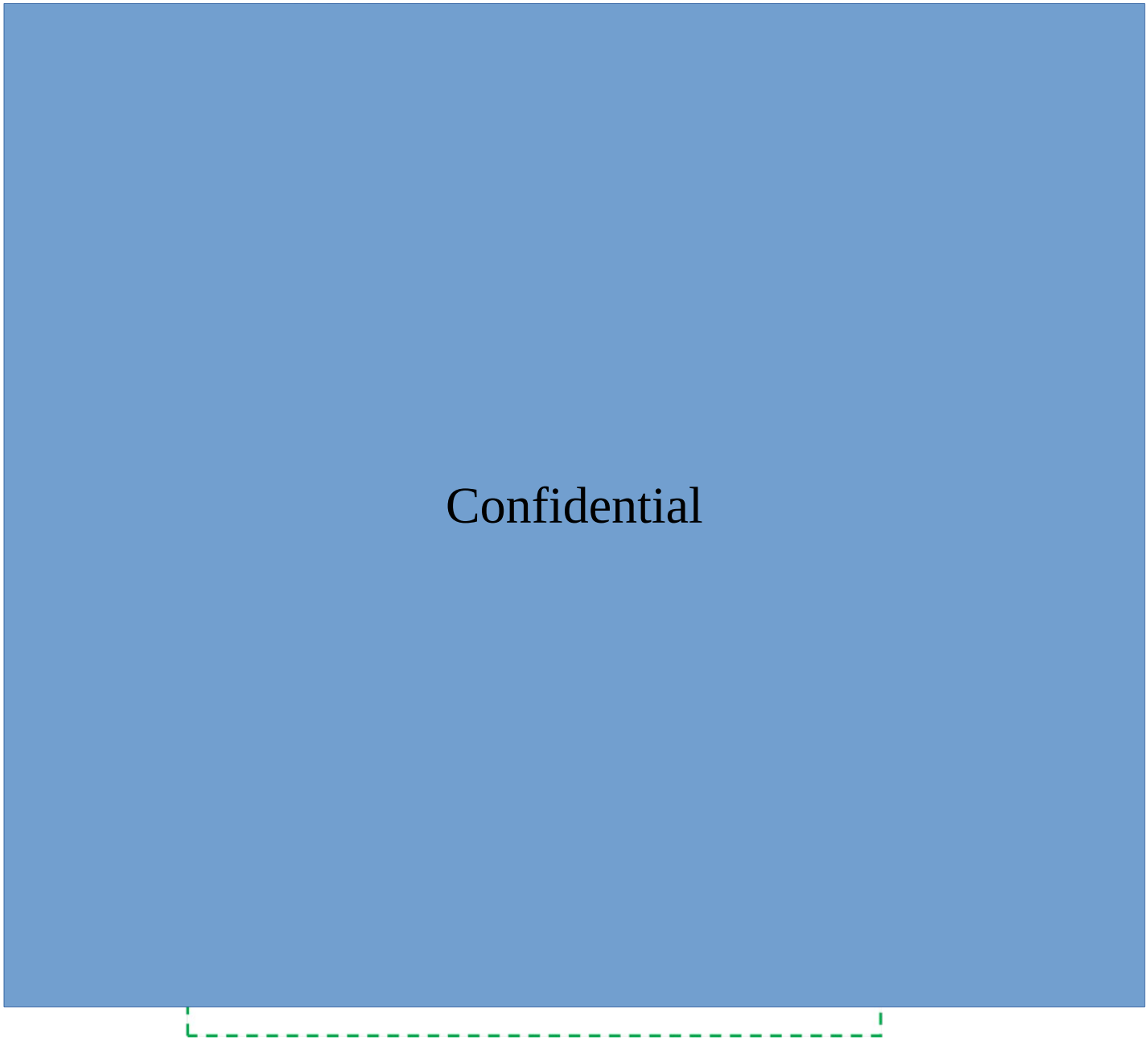


Figure 11: Grid

elevated voltages in the system. The AO algorithm is the second most effective, followed by the PSO algorithm.

The average voltage profile is crucial to ensure the reliability and quality of the power supplied by the PV system. High voltages are necessary to prevent overloads and network disturbances. The average voltage profile of the PSO algorithm is the lowest as the algorithm is not optimized for maintaining high voltages. The PSO algorithm is more likely to place PV panels in locations resulting in low voltages.

The average voltage profile of the AO algorithm is higher than that of the PSO algorithm because the algorithm is optimized to maintain high voltages. The AO algorithm is less likely to place PV panels in locations resulting in low voltages.

The average voltage profile of the WHO algorithm is the highest because the algorithm is optimized to minimize power loss. Minimizing power loss contributes to maintaining high voltages in the system.

In conclusion, the WHO algorithm is the best algorithm for PV placement under scenario 2. It provides the highest

Confidential

Figure 12: System all scenarios

average voltage profile, ensuring the reliability and quality of the power supplied by the system.

B.3. Scenario 3

The table provides the outcomes of placing photovoltaic (PV) panels on a simulated 295-bus system under scenario 3.
It compares the results of four PV placement algorithms PSO, SMA, AO, and WHO,in terms of cost, installed power, power loss, average power factor, and power density.

Cost : The table indicates that the lowest cost is achieved by the SMA algorithm, followed by AO, WHO, and PSO.

Installed Power : Installed power is a critical measure of placement success. The table reveals that the highest installed power is obtained by the PSO algorithm, followed by SMA, AO, and WHO.

Power Loss : Power loss serves as an efficiency gauge for PV placement. The table shows that the least power loss is associated with the WHO algorithm, followed by AO, SMA, and PSO.

Power Density : Power density measures the installed PV power per unit of surface area. The table demonstrates that the highest power density is achieved by the PSO algorithm, followed by SMA, AO, and WHO.

In summary, the WHO algorithm emerges as the optimal choice for PV placement under scenario 3, exhibiting the lowest power loss and the highest average power factor. The SMA algorithm also stands out, offering a well-balanced performance in terms of cost, power loss, and power factor.

The graph illustrates that the WHO algorithm converges most rapidly towards an optimal solution. The PSO algorithm follows, trailed by the AO algorithm.

Swift convergence is crucial for PV placement algorithms, enabling the discovery of an optimal solution within a reasonable timeframe.

The slow convergence of the PSO algorithm is attributed to its reliance on the initial position of particles. If the particles are initially poorly positioned, the algorithm may take more time to converge.

In contrast, the AO algorithm converges faster than the PSO algorithm because it is not dependent on the initial position of chromosomes. The WHO algorithm exhibits the fastest convergence as it employs a direct approach to finding the optimal solution.

In conclusion, the WHO algorithm is the best for PV placement under scenario 3. It converges most rapidly towards an optimal solution, facilitating the discovery of an optimal solution within a reasonable timeframe

In conclusion for this project, we studied PV solutions for improving the performance of interconnected microgrids. First, we made a general state of the art through different themes:

- Renewable power resources
- Interconnected microgrids/smart grid
- Voltage profile improvement
- Power losses reduction
- Power network performance

This allowed us to discover many methods to improve the performance of its microgrids in the field of renewable energy. Then, we were interested and different algorithms that can achieve better performance in total we studied four:

- Particle sawrm optimization
- Wild Horse Otimization
- Aquila Optimization
- Slime Mould Algorithm

Then, in our results we studied algorithms that allow us to achieve the best performance. It depended on the scenarios. Finally, we were able to simulate the different algorithms with the Matlab/Simulink software. In other words, during this semester the project has been completed and we have increased our skills..

References

[1] "Modeling and control of a photovoltaic-wind hybrid microgrid system using ga-anfis," *Heliyon*, vol. 9, no. 4, p. e14678, 2023.

[2] "A bi-level optimization method for voltage control in distribution networks using batteries and smart inverters with high wind and photovoltaic penetrations," *International Journal of Electrical Power Energy Systems*, vol. 151, p. 109217, 2023.

[3] "Analysis and simulation of an urban-industrial sustainable energy community: A use case in san juan de mozarrifar using photovoltaic energy," *Energy Reports*, vol. 9, pp. 1589–1605, 2023.

[4] "The economics of deploying distributed solar photovoltaics in developing countries: Some insights from an analysis for bangladesh," *Energy for Sustainable Development*, vol. 70, pp. 247–258, 2022.

[5] G. Gross, "Contributions of renewable energy resources to resource diversity," pp. 1 pp.–, 2006.

[6] J. T. Bialasiewicz, "Renewable energy systems with photovoltaic power generators: Operation and modeling," *IEEE Transactions on Industrial Electronics*, vol. 55, no. 7, pp. 2752–2758, 2008.

[7] J. Sun, M. Li, Z. Zhang, T. Xu, J. He, H. Wang, and G. Li, "Renewable energy transmission by hvdc across the continent: system challenges and opportunities," *CSEE Journal of Power and Energy Systems*, vol. 3, no. 4, pp. 353–364, 2017.

[8] "Segmentation of a conventional medium voltage network into solar pv powered microgrid," *Energy Reports*, vol. 9, pp. 5183–5195, 2023.

[9] "Transitioning from diesel backup generators to pv-plus-storage microgrids in california public buildings," *Sustainable Production and Consumption*, vol. 38, pp. 252–265, 2023.

[10] "Control techniques for operation of roof-top solar photovoltaics based microgrid in islanded mode," *International Journal of Electrical Power Energy Systems*, vol. 155, p. 109511, 2024.

[11] "Model predictive control based autonomous dc microgrid integrated with solar photovoltaic system and composite energy storage," *Sustainable Energy Technologies and Assessments*, vol. 54, p. 102862, 2022.

[12] "Distributed cooperative voltage mode control for dc-isolated microgrids powered by renewable energy sources," *International Journal of Electrical Power Energy Systems*, vol. 152, p. 109175, 2023.

[13] "Probabilistic optimal planning of multiple photovoltaics and battery energy storage systems in distribution networks: A boosted equilibrium optimizer with time-variant load models," *Journal of Energy Storage*, vol. 73, p. 108986, 2023.

[14] N. T. Mbungu, A. A. Ismail, M. AlShabi, R. C. Bansal, A. Elnady, and A. K. Hamid, "Control and estimation techniques applied to smart microgrids: A review," *Renewable and Sustainable Energy Reviews*, vol. 179, p. 113251, 2023.

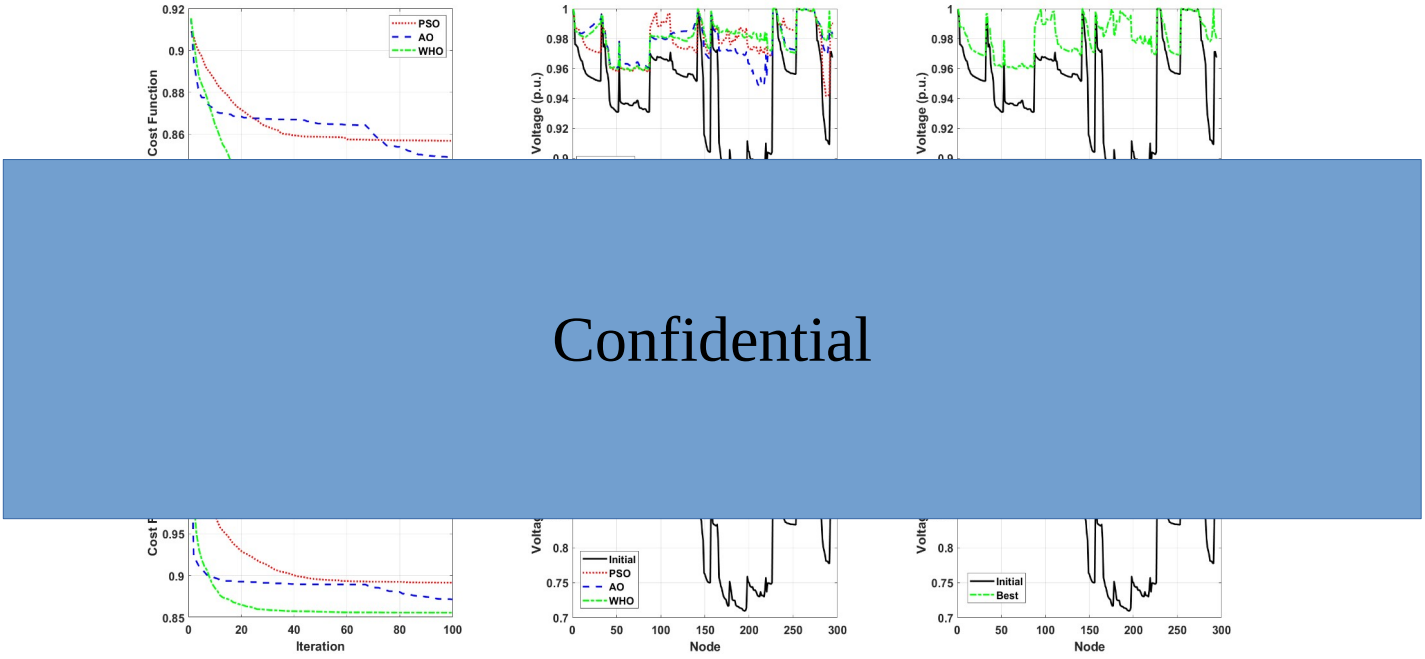
[15] R. Minciardi and M. Robba, "A bilevel approach for the stochastic optimal operation of interconnected microgrids," *IEEE Transactions on Automation Science and Engineering*, vol. 14, no. 2, pp. 482–493, 2017.

[16] S. Rehimi, R. Mirzaei, and H. Bevrani, "Interconnected microgrids frequency response model: An inertia-based approach," *Energy Reports*, vol. 6, pp. 179–186, 2020. The 6th International Conference on Power and Energy Systems Engineering.

[17] S. Mansouri, A. Ahmarinejad, E. Nematbakhsh, M. Javadi, A. Esmaeel Nezhad, and J. CatalÃ£o, "A sustainable framework for multi-microgrids energy management in automated distribution network by considering smart homes and high penetration of renewable energy resources," *Energy*, vol. 245, p. 123228, 2022.

Table 1
Results for the synthetic 295 bus system under scenario 1

	Variable	PSO	SMA	AO	WHO	Best
	Cost	0.857	0.867	0.849	0.823	0.816
PV	TS	(88,176), (79,262)	(104,205),(79,273)	(24,142), (77,228)	(2,208), (47,227)	(2,208),(48,227)



(a) Convergence curve (b) Average Voltage profile (c) Best Voltage profile

Figure 14: Results of best three optimizers for scenario 2.

[18] R. Elazab, A. T. Abdelnaby, H. Keshta, and A. Ali, "Optimal techno-economic feasibility analysis of a grid-tied microgrid considering demand response strategy," *Electric Power Systems Research*, vol. 224, p. 109768, 2023.

[19] "Optimal capacitor placement in northern region of bangladesh transmission network for voltage profile improvement," *Energy Reports*, vol. 9, pp. 1896–1909, 2023.

[20] J. Vanishree and V. Ramesh, "Voltage profile improvement in power systems - a review," pp. 1–4, 2014.

[21] P. Gupta and Y. Pal Verma, "Voltage profile improvement using demand side management in distribution networks under frequency linked pricing regime," *Applied Energy*, vol. 295, p. 117053, 2021.

[22] P. M. Kishore and B. Ravikumar, "Refined hybrid microgrid architecture for the improvement of voltage profile," *Energy Procedia*, vol. 90, pp. 645–654, 2016. 5th International Conference on Advances in Energy Research (ICAER) 2015.

[23] W. J. Praiselin and J. B. Edward, "Voltage profile improvement of solar pv grid connected inverter with micro grid operation using pi controller," *Energy Procedia*, vol. 117, pp. 104–111, 2017. "First International Conference on Power Engineering Computing and CONtrol (PECCON-2017) 2nd -4th March .2017." Organized by School of Electrical Engineering, VIT University, Chennai, Tamil Nadu, India.

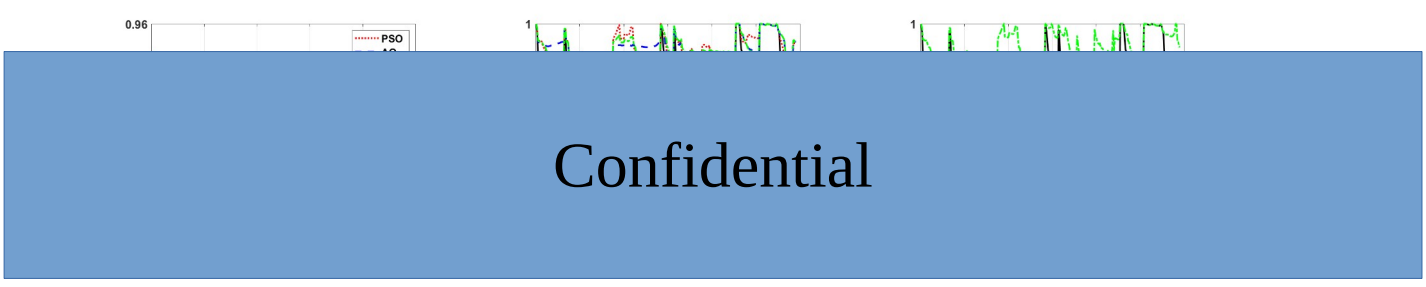
[24] "Implementation of genetic and particle swarm optimization algorithm for voltage profile improvement and loss reduction using capacitors in 132 kv manipur transmission system," *Energy Reports*, vol. 9, pp. 738–746, 2023. 2022 9th International Conference on Power and Energy Systems Engineering.

[25] "Performance enhancement of a photovoltaic thermal system through two-phase nanofluid simulation in a channel equipped with novel artificial roughness," *Applied Thermal Engineering*, vol. 230, p. 120709, 2023.

[26] A. Moreira, Y. Molina, R. Aquino, and Z. Ãaupari, "Allocation and sizing of photovoltaic systems to reduce power losses and economic aspects using a new pso approach," *IEEE Latin America Transactions*, vol. 20, no. 6, pp. 977–985, 2022.

[27] P. A. Afsher, M. V. Manoj Kumar, C. M. Nirmal Mukundan, and K. Shyju, "A pv-dstatcom with adaptive dc-link voltage for grid integration and pq enhancement," *IEEE Transactions on Industry Applications*, vol. 58, no. 5, pp. 6471–6484, 2022.

[28] S. Singh, S. Kewat, B. Singh, B. K. Panigrahi, and M. K. Kushwaha, "A multifunctional three-phase grid coupled solar pv energy conversion system using delayed x00b5;-law proportionate control for pq improvement," *IEEE Transactions on Industry Applications*, vol. 58,



(a) Convergence curve (b) Average Voltage profile (c) Best Voltage profile

Figure 15: Results of best three optimizers for scenario 3.

no. 1, pp. 554–564, 2022.

[29] F. Chen, H. Huang, J. Wu, W. Xu, B. Wang, and H. Lu, "Research on potential power loss reduction of distribution line," in *The 16th IET International Conference on AC and DC Power Transmission (ACDC 2020)*, vol. 2020, pp. 976–979, 2020.

[30] J. Shi, X. Pan, F. Xu, and S. Ye, "Research on multi-dimensional early warning system under the integrated display of precise loss reduction based on intelligent management and control of loss reduction," in *2022 International Conference on Electronics and Devices, Computational Science (ICEDCS)*, pp. 337–341, 2022.

[31] T. T. Nguyen, T. T. Nguyen, and H. P. Nguyen, "Optimal soft open point placement and open switch position selection simultaneously for power loss reduction on the electric distribution network," *Expert Systems with Applications*, vol. 238, p. 121743, 2024.

[32] L. Asiminoaei, P. Rodriguez, F. Blaabjerg, and M. Malinowski, "Reduction of switching losses in active power filters with a new generalized discontinuous-pwm strategy," *IEEE Transactions on Industrial Electronics*, vol. 55, no. 1, pp. 467–471, 2008.

[33] J. Olivares, Y. Liu, J. Canedo, R. Escarela-Perez, J. Driesen, and P. Moreno, "Reducing losses in distribution transformers," *IEEE Transactions on Power Delivery*, vol. 18, no. 3, pp. 821–826, 2003.

[34] J. Zhu, K. Cheung, D. Hwang, and A. Sadjadpour, "Operation strategy for improving voltage profile and reducing system loss," *IEEE Transactions on Power Delivery*, vol. 25, no. 1, pp. 390–397, 2010.

[35] B. Long, D. Shen, T. Cao, J. Rodriguez, C. Garcia, J. M. Guerrero, and K. To Chong, "Power losses reduction of t-type grid-connected converters based on tolerant sequential model predictive control," *IEEE Transactions on Power Electronics*, vol. 37, no. 8, pp. 9089–9103, 2022.

[36] J. A. Momoh, K. D'Arnaud, A. Chuku, and F. Posey-Eddy, "Performance optimization of power supply networks using renewable energy resources," in *41st North American Power Symposium*, pp. 1–6, 2009.

[37] Y. Linlin, Z. Lihua, M. Gaojun, Z. Feng, and L. Wanxun, "Research on multi-objective reactive power optimization of power grid with high proportion of new energy," *IEEE Access*, vol. 10, pp. 116443–116452, 2022.

[38] "Real-time coordinated control of voltage regulation devices in a high pv penetrated weak distribution network," *Computers and Electrical Engineering*, vol. 110, p. 108872, 2023.

[39] "Hvdc grid fault location method using genetic algorithm on reconstructed frequency-domain voltage profiles," *International Journal of Electrical Power Energy Systems*, vol. 144, p. 108429, 2023.

[40] Y. Linlin, Z. Lihua, M. Gaojun, Z. Feng, and L. Wanxun, "Research on multi-objective reactive power optimization of power grid with high proportion of new energy," *IEEE Access*, vol. 10, pp. 116443–116452, 2022.

[41] "Fuzzy risk-based framework for scheduling of energy storage systems in photovoltaic-rich networks," *Journal of Energy Storage*, vol. 52, p. 104902, 2022.

[42] "Low-breakdown-voltage solar cells for shading-tolerant photovoltaic modules," *Cell Reports Physical Science*, vol. 3, no. 12, p. 101155, 2022.

[43] "Data-driven voltage/var optimization control for active distribution network considering pv inverter reliability," *Electric Power Systems Research*, vol. 224, p. 109800, 2023.

[44] C. D. Sanjenbam, B. Singh, and P. Shah, "Reduced voltage sensors based upqc tied solar pv system enabling power quality improvement," *IEEE Transactions on Energy Conversion*, vol. 38, no. 1, pp. 392–403, 2023.

[45] S. B. Q. Naqvi and B. Singh, "An enhanced filtering generalized integrator-based control for improved performance of a grid-tied pv system at adverse grid voltages," *IEEE Transactions on Industrial Electronics*, vol. 70, no. 12, pp. 12376–12386, 2023.

[46] D. Prasad and D. C., "Solar pv-fed multilevel inverter with series compensator for power quality improvement in grid-connected systems," *IEEE Access*, vol. 10, pp. 81203–81219, 2022.

[47] E. A. Mohamed, M. Aly, A. Elmelegi, E. M. Ahmed, M. Watanabe, and S. M. Said, "Enhancement the frequency stability and protection of interconnected microgrid systems using advanced hybrid fractional order controller," *IEEE Access*, vol. 10, pp. 111936–111961, 2022.

[48] Q. Jin and L. Sun, "Microgrid operation optimization based on wild horse optimizer," pp. 937–942, 2023.

[49] M. Alshareef, Z. Lin, M. Ma, and W. Cao, "Accelerated particle swarm optimization for photovoltaic maximum power point tracking under partial shading conditions," *Energies*, vol. 12, p. 623, 02 2019.

[50] R. Pereira de Gusmao and F. de Assis Tenorio de Carvalho, "Particle swarm optimization applied to relational data clustering," in *2016 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 001690–001695, 2016.

[51] W. Aribowo, H. Suryoatmojo, and F. A. Pamuji, "Optimalization droop control based on aquila optimizer algorithm for dc microgrid," in *2022 International Seminar on Intelligent Technology and Its Applications (ISITIA)*, pp. 460–465, 2022.